MASTER DATA

A visual guide to the five data domains, every supported upload field, and how master records feed a tenant MRP plan.

IMPORTANT: SAMPLE DISPLAY VS. REAL TENANT DATA

The Master Data grid and its counts are currently hard-coded demonstration rows. CSV upload is a separate working flow that writes supported records into the selected tenant database. The grid does not yet display uploaded records.

DEMO GRID != DATABASE CATALOG

THE PLANNING FLOW

Plant | **Materials** | **Products + BOM** | **Constraints** | **Stock + demand**

Then MRP explodes demand through the BOM, nets it against stock and receipts, and applies safety stock, order sizing, lead time, and capacity checks.

01 / READ THE PAGE CORRECTLY

Screen map

Understand which controls are operational before relying on the displayed values.

DATABASE-BACKED

Working tenant path

Select the tenant, select an upload type, and upload a valid CSV. The server validates fields and references, then writes records to that tenant schema.

FRONTEND SAMPLE

Demo display path

Grid rows, counts, owners, statuses, KPIs, sync time, data score, Add/Edit, Toggle and Export are local sample UI, not persisted tenant data.

HEADER + CONTROLS

CONTROL	WHAT IT DOES
Tenant selector	Sets tenant context for CSV API uploads; verify before every upload.
Export	Downloads filtered demo rows; it is not a database export.
Upload type	Plant master, Material master, Finished goods, Plant constraints, or BOM.
Upload CSV	Uploads selected CSV and reports success or validation/reference errors.
Add [record]	Opens generic demo form; browser-only state, not a database create.
Search / status	Filters displayed demo rows by text and sample status.
Row actions	Edit/Toggle are local; Copy code copies the displayed identifier.
Checkbox / pages	Visual sample controls; no connected bulk or database pagination.

GRID COLUMNS

FIELD	MEANING
Code / Name	Business identifier and readable label for the sample row.
Category / Type	Grouping whose meaning changes with the selected domain.
Unit	UOM for items; demo table may reuse it for lead time, location, or value.
Planning attributes	Short demo note; real import uses individual typed CSV columns.
Updated / Owner / Status	Illustrative labels, not tenant audit or activation state.
Master records / Active	Counts over local demo arrays, not uploaded tenant rows.
Needs review / Last sync	Static demonstration metrics; no live quality scan or sync timestamp.
Data quality	Fixed sample percentage; not computed from database validation.

ADD / EDIT FORM

Code and Name are required. Category/Type, Unit or Value, and Planning attributes/Notes are generic optional inputs. They do not map all database columns and remain local demo state.

02 / THE FIVE DATA DOMAINS

Purpose, source, and use

These are page groupings, not five physical database tables.

1 / MATERIALS

Components and supply items

Demo examples: RM-0045 Aluminum Sheet, MT-0012 Motor Assembly, PK-3011 Packaging Box, SP-7781 Screw M6, RM-1120 Copper Wire, EL-2204 Control Relay.

Actual MRP source: Material master CSV. Defines unit, cost, scrap, sourcing and replenishment policy.

2 / PRODUCTS + BOMS

Forecastable products and recipes

Demo products: PRD-1001 Control Panel, PRD-1002 Switch Assembly, PRD-1003 Power Module, PRD-1004 Enclosure Unit.

Actual MRP sources: Finished goods + Bill of materials CSVs. BOM ratios translate product demand into components.

3 / SUPPLIERS

Vendor identity and performance

Demo examples: SUP-001 ABC Suppliers, SUP-002 Global Traders, SUP-004 PackRight Solutions.

Current state: hard-coded sample only. No Supplier upload or database CRUD path exists on this page.

4 / LOCATIONS

Sites, plants and warehouses

Demo examples: LOC-001 Main Warehouse, LOC-002 Plant North, LOC-003 Plant Central.

Actual MRP source: Plant master CSV. Plant code scopes demand, stock, products, BOM and constraints.

5 / PLANNING PARAMETERS

Replenishment policy values

Demo examples: Default service level, Planning time fence, Default calendar.

Actual MRP sources: Material master + Plant constraints. The demo items themselves are not applied.

OTHER PLANNING INPUTS

Stock and forecast

These are not Master Data tabs. Inventory and Demand Planning provide changing, time-phased operating data.

Actual sources: Inventory CSV + weekly/monthly demand CSV or imported workbook scenario.

WHERE DO THE VISIBLE RECORDS COME FROM?

The example codes and names above are seeded in the frontend source for screen demonstration. They do not come from SAP, ERP, a vendor integration, or the selected tenant database. A CSV upload may succeed even though the static grid does not refresh.

DOMAIN TO REAL UPLOAD TYPE

PAGE DOMAIN	TENANT DATABASE INPUT
Materials	Material master
Products	Finished goods + Bill of materials
Locations	Plant master
Planning Parameters	Material master + Plant constraints
Suppliers	No upload type currently supported

03 / CSV FIELD REFERENCE

Plant + material master

Upload these foundation records first; later files depend on their identifiers.

PLANT MASTER

Answers: where does planning take place? Country code and plant code are shared references for tenant operations.

CSV FIELD	MEANING + MRP USE
plant_code	Stable plant key referenced by products, BOM, inventory, constraints, and demand.
plant_name	Readable name for the facility.
country_code	Three-letter country code for country filter and currency context; country must be configured or supplied with supported metadata.
location	Descriptive city, region, or site location.
active_flag	true/false; inactive plants are excluded from planning options.

MATERIAL MASTER

Answers: what item is needed, in what unit, and with what supply and replenishment policy?

CSV FIELD	MEANING + MRP USE
material_code	Unique tenant item key referenced by BOM, inventory, and constraints.
description	Readable material name.
category	Raw, Packing, or Intermediate; classification and filtering.
supply_type	BUY = procure externally. MAKE = manufacture internally; MAKE materials may have their own BOM.
uom	Stock/planning unit such as ton, kg, or each. Keep coherent with BOM and stock.
unit_cost	Cost per unit used in planned procurement value and spend reporting.
scrap_pct	Loss fraction from 0 up to but not including 1; 0.02 means 2% extra consumption.
lead_time_buckets	Weekly/monthly replenishment buckets; interpretation follows the run time-bucket type.
moq	Minimum order quantity applied when net requirement is positive.
order_multiple	Order increment to round suggested quantity upward to.
safety_stock_min	Default projected-stock floor; plant constraints may override it.
max_capacity	Default capacity ceiling for alerts; plant constraints may override it.

Keep units consistent: BOM consumption and stock must use compatible UOMs.

04 / CSV FIELD REFERENCE

Products + replenishment policy

These records connect forecasted finished goods to the materials that must be supplied.

FINISHED GOODS

CSV FIELD	MEANING + MRP USE
plant_code	Plant where product is planned; must exist.
fg_code	Finished-good key used by demand and BOM parent.
description	Readable product name.
product_type	Bulk or Bagged; supports product-type filtering.
uom	Forecast/output unit coherent with demand and BOM ratio.
active_flag	true/false; only active products are available for planning.

BILL OF MATERIALS (BOM)

CSV FIELD	MEANING + MRP USE
plant_code	Plant-specific product structure.
fg_code	Parent finished good; must exist at the same plant.
component_material_code	Consumed child material; must exist in Material master.
quantity_per_fg	Component quantity needed per one parent unit; core BOM ratio.
quantity_uom	Source unit for component quantity; keep consistent with material.
scrap_factor	Optional source field. Current import/calculation ignores it; material scrap_pct is used.

BOM upload replaces all existing components for every parent included.

Upload complete component rows; a partial file removes omitted components.

PLANT CONSTRAINTS

CSV FIELD	MEANING + MRP USE
plant_code + material_code	Plant/material pair the rule applies to; both must exist.
min_stock_qty	Plant-level projected stock floor used as safety-stock target.
reorder_level	Stored reorder point; not used by current forward netting formula.
max_stock_qty	Maximum stock reference and fallback capacity limit.
max_storage_capacity	Storage limit used for capacity alert when provided.
lead_time_buckets	Plant/material lead-time override; material default is fallback.
moq	Plant/material minimum-order override.
order_multiple	Plant/material order-increment override.
uom	Optional reference unit for constraint quantities.

Demo service level, planning time fence, and calendar are not currently applied by forward MRP.

05 / FROM UPLOAD TO DECISION

How MRP uses master data

The order below satisfies reference checks and produces traceable time-phased requirements.

1 PLANT site + country	2 MATERIALS items + policy	3 PRODUCTS finished goods	4 CONSTRAINTS plant overrides
5 BOM complete recipe	6 STOCK on-hand + receipts	7 DEMAND forecast periods	8 RUN MRP selected tenant

GROSS COMPONENT REQUIREMENT

Forecast x BOM ratio x (1 + material scrap_pct)

Aggregated by plant, material, and planning time bucket.

CALCULATION

NET + SIZE

Net scheduled and in-transit receipts and gross requirements against projected stock. If stock would fall below the minimum, the gap is net requirement. Apply MOQ and order multiple; use lead time for release timing.

OUTPUT

ALERT + REVIEW

Capacity alert when projected stock exceeds configured capacity. Expedite alert when release is due by the first bucket. MRP creates recommendations; it does not send a supplier order.

RECOMMENDED UPLOAD ORDER

STEP	ACTION
1	Plant master
2	Material master
3	Finished goods
4	Plant constraints
5	Bill of materials
6	Inventory & stock
7	Demand / scenario
8	Run MRP

CURRENT FORMULA BOUNDARIES

Reserved stock is not deducted; reorder_level is stored but not netted; BOM scrap_factor is ignored. Material scrap_pct is applied. Maximum capacity flags excess stock but does not cap order quantity.